SAHIL LENKA

Full-Stack Developer | Python, React, Node.js | Open Source Contributor
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SUMMARY

Second-year CS student (AI & ML) building full-stack and backend software in Python and JavaScript. Merged 40+ pull requests across Concore, Beehive, Diomede and Krkn (CNCF Sandbox), and published ws-tap on npm. Interned on a production website at IOTA Studio AI and on Razorpay payments at Sundarone.

EDUCATION

Manipal University Jaipur **Aug 2024 – May 2028**
B.Tech. in Computer Science & Engineering (AI & ML) *Jaipur, India*

OPEN SOURCE CONTRIBUTIONS

Concore – ControlCore-Project | Python, C++, ZeroMQ, Docker **15 merged PRs**

- Tracked down a race where two writers could **corrupt the same shared-memory message**, and fixed it with a sequence counter and a POSIX semaphore in Python and C++. Backed it with 3 regression tests and a C++ CI job.
- Wrote the **concore CLI from scratch**: 6 commands (init, run, validate, inspect, status, stop), with unit tests and GitHub Actions CI.

Beehive & Diomede – KathiraveluLab | Python, Flask, Redis, pynetdicom **22 merged PRs**

- Built a static scanner that hunts for API routes missing auth. It found **6 vulnerabilities (1 critical, 4 high)**. Also locked down secret handling and turned off debug mode in production.
- Built **load-based DICOM routing** across multiple Orthanc servers, then made it fail gracefully: a clean HTTP 503 when Redis is down and retries for failed acknowledgements.

Krkn – krkn-chaos, **CNCF Sandbox** | Python, Kubernetes **5 merged PRs**

- Plugged a **thread-pool leak** that left worker threads running after cloud API errors, with 2 tests to keep it fixed.

EXPERIENCE

Sundarone Private Limited **May 2026 – Aug 2026**
Software Development Intern *India*

- Added Razorpay payments to the booking flow, which helped lift bookings **15%** after launch.
- Worked with the team on rounds of UX changes to the booking pages that helped lift retention **20%**.

IOTA Studio AI **Apr 2025 – Aug 2025**
Web Development Intern *Remote*

- Built and shipped the company's **production website end to end** with React.js, Node.js and REST APIs, landing 10+ commits in production.
- Designed the **REST API** between the React frontend and the Node.js backend, with content in MongoDB and SQL.

PROJECTS

ws-tap – published on npm | JavaScript, Node.js, Socket.IO **2026**

- Tired of sprinkling console.log through Socket.IO handlers, I built **one line of middleware** that streams every event to a live terminal viewer with pause and filters. Zero dependencies, about 200 lines.
- Redacts passwords, tokens and auth headers **before anything leaves the process**. 11 tests across Linux, Windows and macOS.
- **Safe to leave in**: switches itself off in production, swallows its own errors, and never keeps your process alive.

EvidenceRAG | Python, FastAPI, Ollama **2026**

- Built a local-first document Q&A API that would rather **say "I don't know" than guess**: an NLI model has to confirm every cited passage. 97 backend tests.
- Mixed dense and BM25 search with rank fusion and cross-encoder reranking. On my **20-question test set** it ranks the right passage first every time (MRR 1.000, vs 0.975 for dense search alone).
- Fingerprints documents with **SHA-256** so unchanged files never get re-embedded, and streams answers over Server-Sent Events.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, SQL
Backend: FastAPI, Flask, Node.js, REST APIs, Socket.IO, webhooks
Data & Infrastructure: MongoDB, Redis, Docker, GitHub Actions CI/CD, Linux
Testing & Retrieval: pytest, node:test, RAG, BM25, cross-encoder reranking, Ollama

ACHIEVEMENTS

- **HackRx 6.0 (Bajaj Finserv):** Top 35 of 10,000+ teams | **HackOn with Amazon 6.0:** Grand Finalist
- **Smart Delhi Ideathon:** 3rd place